

Question 1 (10 marks)

- (a) Simplify $2\sqrt{32} + \sqrt{20} - 4\sqrt{18}$ 2
- (b) Evaluate correct to 3 significant figures 2
- $$\frac{\sqrt[3]{632.15}}{18.27+6.3}$$
- (c) Write the following as a single fraction with rational denominator 2
- $$\frac{1}{2\sqrt{2}} - \frac{3+\sqrt{2}}{3\sqrt{2}}$$
- (d) Express $0.\dot{3}4\dot{2}$ as a fraction in its simplest form. Showing all your working. 2
- (e) Given that $F = \frac{9C}{5} + 32$.
- (i) Find the value of F when $C = 25$. 1
- (ii) Make C the subject of the formula. 1

Question 2 (Start a new page) (12 marks)

- (a) Factorise the following
- (i) $x^2 - 10x + 21$ 1
- (ii) $2a^2 + 9ab - 5b^2$ 2
- (iii) $4a^2 - 4a + 8ab - 8b$ 2
- (b) Solve the following equations
- (i) $\frac{x+3}{2} - \frac{3x-1}{7} = 2$ 2
- (ii) $x + \frac{1}{x} = 3$ 2
- (iii) $|3 - 2x| = 5x - 3$ 3
- (c) Solve the following inequality and graph the solution on the number line
- $$|3x - 2| \geq 1$$
- 2

Question 3 (Start a new page) (12 marks)

- (a) By rationalising the denominator find the value of a and b 2

$$\frac{8}{3 - \sqrt{5}} = a + b\sqrt{5}$$

- (b) Find the domain and range of the following: 4

(i) $y = 9 - x^2$

(ii) $y = \frac{1}{\sqrt{8-4x}}$

- (c) $f(x) = \begin{cases} 5 - x^2 & \text{if } x \geq 1 \\ (x - 3)^2 & \text{if } x < 1 \end{cases}$ find the value of $f(3) - 2f(-6)$ 2

- (d) Determine if $f(x) = \frac{2x}{2x^2 - 1}$ is an odd function, even function or neither. 2

- (e) Solve the equation $2\cos\theta = -\sqrt{3}$ where $0 \leq \theta \leq 360^\circ$ 2

Question 4 (Start a new page) (12 marks)

- (a) If $\cos\theta = -0.6$ and $\sin\theta < 0$, find the exact value of $\tan\theta$. 2

- (b) Simplify $\frac{2}{x(3-x)} - \frac{1}{x}$ 2

- (c) On a number plane shade in the region given by the two inequalities 3

$$y > x^2 - 2 \text{ and } y \leq x. \text{ (Indicate where each graph crosses the } x \text{ and } y \text{ axes)}$$

- (d) Solve for x , if $9^x - 4(3^x) + 3 = 0$ 3

- (e) Solve for x and y 2

$$\begin{aligned} 2x + 3y &= -1 \\ -5x + 2y &= -7 \end{aligned}$$

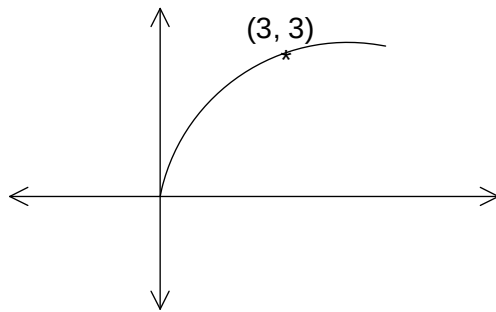
Question 5 (Start a new page) (10 marks)

(a) Solve $\sin^2 x - 2\cos x + 2 = 0$ for $0^\circ \leq x \leq 360^\circ$ 3

(b) Find the exact value of $\frac{\sin^2 60^\circ}{\cot 60^\circ + \sec 30^\circ}$ 2

(c) In 1 hour 80 people attend a swimming pool and the money taken at the gate amounts to \$139. If adults are charged \$2.50 and children are \$1.50, how many of each entered the pool that hour? 2

(d) The diagram represents a portion of an even function $y = f(x)$ 2

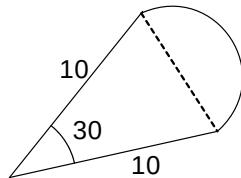


(i) Redraw this diagram and complete the graph .

(ii) Find $f(3) + f(-3)$

Question 6 (Start a new page) (10 marks)

(a) A sector is given below. Find the area of the shaded region. 2



(b) Prove $(\operatorname{cosec} \beta + \cot \beta)(\operatorname{cosec} \beta - \cot \beta) = 1$ 2

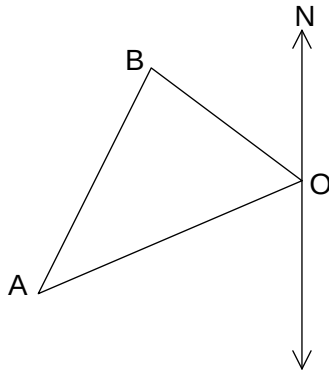
(c) The function $f(x)$ is defined as $f(x) = \begin{cases} 2x & -4 \leq x < 0 \\ 9 - x^2 & 0 \leq x \leq 3 \end{cases}$

(i) Sketch $y = f(x)$ 2

(ii) State the range of $y = f(x)$ 1

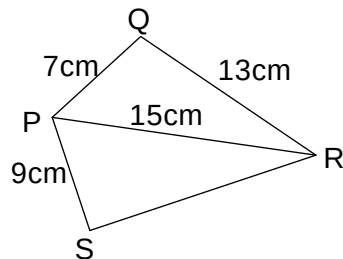
Question 6 – Cont

- (d) From an observation tower O, a man determines the lighthouse A is on a bearing of 225° T and another lighthouse B is at a bearing of 315° T. Given that A and B are at a distance of 75 and 70 nautical miles respectively from O, find
- (i) the distance between A and B. 2
- (ii) The bearing of B from A to the nearest degree. 1



Question 7(Start a new page) (10 marks)

- (a) For the parabola $y = x^2 - 2x - 3$
- (i) find its x and the y intercepts 1
- (ii) find its vertex and hence draw a neat sketch of the curve. 2
- (b) Show that $\frac{4}{2 + \sqrt{5}} - \frac{1}{9 - 4\sqrt{5}}$ is rational. 2
- (c) Simplify $\frac{1 - \frac{1}{x+1}}{\frac{1}{x+1}}$ 2
- (d) In the diagram $\angle SPQ = 147^\circ$. Let $\angle QPR = \theta$



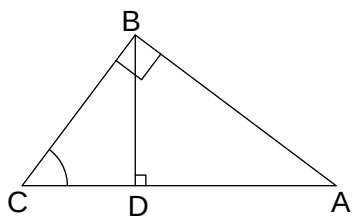
- (i) Show that $\cos \theta = \frac{1}{2}$ 2
- (ii) Find the area of $\triangle PRS$. 1

Question8 (start a new page) (13marks)

- (a) Given $x^2 + 12x = p$, use the method of completing square to find an expression for x in terms of p . If $p = 253$, find all possible values of x . 3
- (b) If $x + y = 1$ and $x^3 + y^3 = 19$ Find the value of $x^2 + y^2$. 2
- (c) (i) Sketch on the same number plane $y = \cos x$ and $y = \sin x$ where $0^\circ \leq x \leq 90^\circ$ 2
- (ii) Find the point of intersection of $y = \cos x$ and $y = \sin x$ for $0^\circ \leq x \leq 90^\circ$ 1
- (iii) Hence or otherwise solve the inequality $\sin x > \cos x$ for $0^\circ \leq x \leq 90^\circ$ 1
- (d) Simplify $\frac{5^{n+3} - 5^n}{5^n}$ 2
- (e) Simplify $\frac{6x^3 + 48}{24 - 6x^2}$ 2

Question9 (Start a new page) (13 marks)

- (a) If $f(x) = 1 - x^2$ and $g(x) = 2x + 1$
- (i) find $f(-2)$ 1
- (ii) find the value of x for which $f(x) = g(x)$ 2
- (iii) find $f[g(x)]$ 1
- (b) $\triangle ABC$ is right-angled at B.
- D is a point on AC so that BD is perpendicular to AC, also $BC = 1$ unit
- $DA = CD + 1$ and $\angle BCA = \theta$
- (i) show that $2\cos\theta + 1 = \sec\theta$. 2
- (ii) Deduce that $2\cos^2\theta + \cos\theta - 1 = 0$. 2
- (iii) Hence find θ . 1



Question 9 – Cont

(c) Find the value of A , B and C if $y^2 + 16y + 94 - 6x$ is expressed in the form 2

$$(y + C)^2 - B(x + A).$$

(d) If $\frac{2}{3x-2c} + \frac{3}{2x-3c} = \frac{7}{2c}$ find x in terms of c . 2

- END OF EXAM -

Solution Yr 11 Half yearly 2011

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21a) $2 \times 4\sqrt{2} + 2\sqrt{5} - 4 \times 3\sqrt{2}$ — ①

$$\left. \begin{array}{l} 8\sqrt{2} + 2\sqrt{5} - 12\sqrt{2} \\ 2\sqrt{5} - 4\sqrt{2} \end{array} \right\} \text{①}$$

b) $0.349302 \dots$ — ①

0.349 — ①

c) $\frac{\sqrt{2}}{4} - \frac{3\sqrt{2}+2}{6}$ — ①

$$\frac{3\sqrt{2}-6\sqrt{2}-2}{12} \text{ — ①}$$

$$= \frac{-3\sqrt{2}-2}{12} \text{ or } -\frac{(3\sqrt{2}+2)}{12}$$

d) let $x = 0.342342 \dots$

① — $1000x = 342.342 \dots$

$$999x = 342$$

$$x = \frac{342}{999} = \frac{38}{111} \text{ — ①}$$

Should be in the simplest form

e) $P = \frac{9C}{5} + 32$

ii) $f = 9 \times \frac{25}{5} + 32 = 77$ — ①

iii) $\frac{9C}{5} = f - 32$

$$9C = 5f - 160$$

$$C = \frac{5f - 160}{9} \text{ — ①}$$

10

22 a) $(x-7)(x-3)$ — ①

ii) $\frac{(2a+10b)(2a-b)}{2}$
 $= (a+5b)(2a-b)$ — ①

iii) $4a(a-1) + 8b(a-1)$
 $(a-1)(4a+8b)$ — ①

$$4(a-1)(a+2b) \text{ — ①}$$

b) i) $7x+21-6x+2 = 28$
 $x+23 = 28$
 $x = 5$ — ②

ii) $x^2+1 = 3x$
 $x^2-3x+1 = 0$

$$x = \frac{3 \pm \sqrt{9-4}}{2}$$

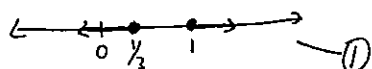
$$= \frac{3 \pm \sqrt{5}}{2} \text{ — ②}$$

iii) $3-2x = 5x-3$ | $3-2x = -5x+3$
 $7x = 6$ | $7x = 0$
 $x = 6/7$ — ① | $x = 0$ — ①

Check — ①

$x = 0$ | $x = 6/7$
LHS $|3-2x| = 3$ | LHS $|3-2 \times 6/7| = 9/7$
RHS = -3 | RHS $5 \times 6/7 - 3 = 9/7$
 $x = 0$ is not a sol. | $\therefore x = 6/7$ is a sol.

g) $3x-2 \geq 1$ | $3x-2 \leq -1$
 $3x \geq 3$ | $3x \leq 1$
 $x \geq 1$ | $x \leq 1/3$



23 a) $\frac{8}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}}$

$$\frac{24+8\sqrt{5}}{9-5} = \frac{24+8\sqrt{5}}{4}$$

$$= 6+2\sqrt{5} = a+b\sqrt{5}$$

$a=6$ | $b=2$
① | ①

b) i) D: all real x — ①

R: $y \leq 9$ — ①

ii) D: $8-4x > 0$
 $x < 2$ — ①

R: $y > 0$ — ①

c) $f(3) - 2f(-6) = 5-3^2 - 2(-9)^2$
 $= -4-162$
 $= -166$ — ②

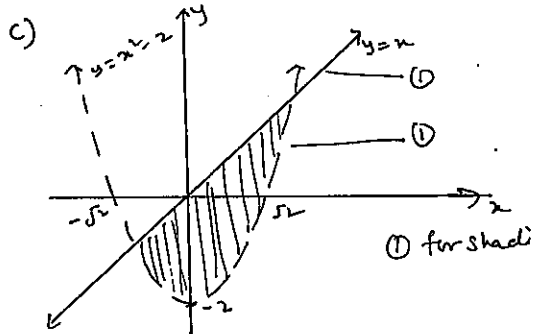
d) $f(x) = \frac{-2x}{2x^2-1} = -f(x)$
Hence $f(x)$ is an odd fn — ②
(with working)

e) $\cos \phi = \frac{-\sqrt{3}}{2}$ | $\frac{v}{v}$
Acute $\angle = 30^\circ$
 $\phi = 150^\circ, 210^\circ$ — ②

12

$\cos \theta = \frac{-6}{10} = \frac{-3}{5}$ |
 $\tan \theta = \frac{4}{3}$ or $\frac{4}{3}i$ — ②

b) $\frac{2-3+x}{x(3-x)} = \frac{x-1}{x(3-x)}$ — ②



d) $3^{2x} - 4(3^x) + 3 = 0$

let $3^x = u$ — ①

$$u^2 - 4u + 3 = 0$$

$$(u-3)(u-1) = 0$$

$u=3, u=1$ — ①

$3^x = 3 \therefore x=1$
 $3^x = 1 \therefore x=0$ — ①

e) $2x+3y = -1 \times 5$
 $-5x+2y = -7 \times 2$
 $10x+15y = -5$
 $-10x+4y = -14$

$$19y = -19$$

$$y = -1$$

$$2x-3 = -1$$

$$2x = 2$$

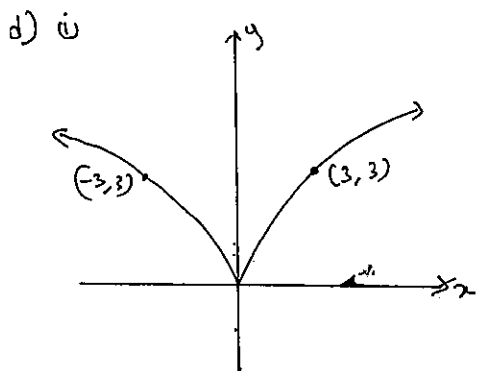
$$x = 1$$

12

35 a) $\sin^2 x - 2 \cos x + 2 = 0$
 $1 - \cos^2 x - 2 \cos x + 2 = 0$ — ①
 $\cos^2 x + 2 \cos x - 3 = 0$
 $(\cos x + 3)(\cos x - 1) = 0$ — ①
 $\cos x \neq -3$
 $\cos x = 1$
 $x = 0^\circ \text{ or } 360^\circ$ } ①

b) $\frac{3}{4}$ — ① $\frac{\sqrt{3}}{4}$ — ①
 $\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}$

c) $2.5a + 1.5c = 138$
 $a + c = 80$
 $5a + 3c = 278$
 $5a + 240 - 3a = 278$
 $2a = 38$
 $a = 19$
 $c = 61$ } ①



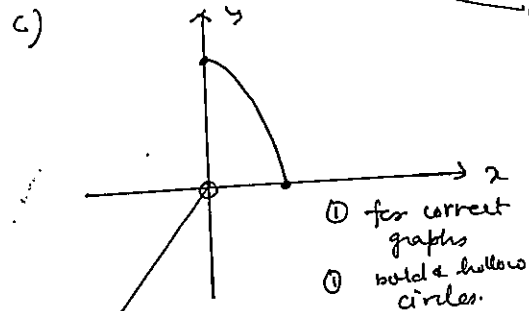
ii) $f(3) + f(-3) = 3 + 3 = 6$ — ①

9

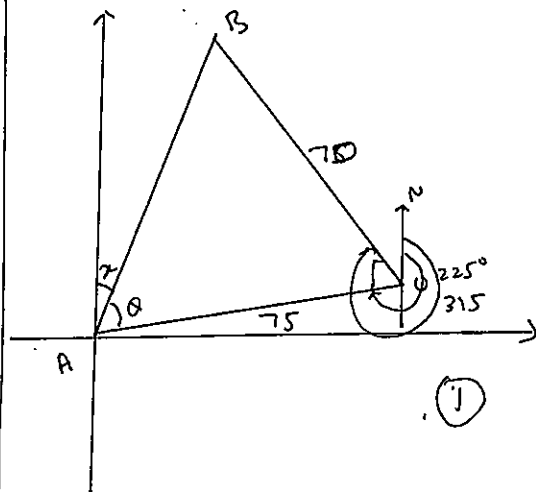
36 a) $A = \text{Area of Sector} - \text{Area of } \Delta$
 $= \frac{1}{2} \pi r^2 \left(\frac{60}{360} \right) - \frac{1}{2} \times 10 \times 10 \times \sin 30^\circ$ — ①
 $= \frac{1}{2} \times \pi \times 10^2 \times \frac{1}{6} - 50 \times \frac{1}{2}$
 $= 50 \left(\frac{\pi}{6} - \frac{1}{2} \right) = 25 \left(\frac{\pi - 3}{3} \right)$ — ①
 (or any equivalent ans)

b) LHS $\left(\frac{1}{\sin \beta} + \frac{\cos \beta}{\sin \beta} \right) \left(\frac{1}{\sin \beta} - \frac{\cos \beta}{\sin \beta} \right)$
 $= \left(\frac{1 + \cos \beta}{\sin \beta} \right) \left(\frac{1 - \cos \beta}{\sin \beta} \right)$
 $= \frac{1 - \cos^2 \beta}{\sin^2 \beta} = \frac{\sin^2 \beta}{\sin^2 \beta} = 1 = \text{RHS}$ — ②

OR
 LHS $\sec^2 \beta - \cot^2 \beta = 1 = \text{RHS}$ — ②



ii) $-8 \leq y \leq 9$ — ①



ii) $\angle AOB = 90^\circ$
 $AO^2 = 70^2 + 75^2$
 $AB = 102.59$ — ①

iii) $\frac{\sin \theta}{70} = \frac{\sin 90}{102.59}$

$\sin \theta = \frac{70}{102.59}$

$\theta = 43^\circ 2'$

$\therefore$ Bearing B from A

$180 = 135 + 43^\circ 2' + x$

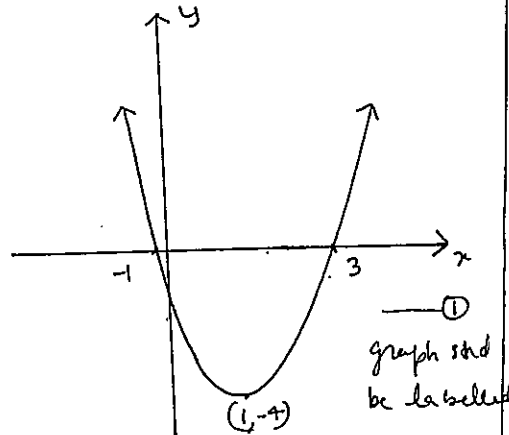
(cont. 2 mil. long)

$= 2^\circ$ — ①

10

37 a) i) $y \text{ int} = 3$
 $x \text{ int} \rightarrow (x-3)(x+1) = 0$ — ①
 $x = 3, -1$

ii) axis of sym is $x = 1$
 $\therefore y \text{ coordinate is}$
 $y = 1 - 2 - 3 = -4$
 $V(1, -4)$ — ①



b) $\frac{4}{2+\sqrt{5}} \times \frac{2-\sqrt{5}}{2-\sqrt{5}} - \frac{1}{9-4\sqrt{5}} \times \frac{9+4\sqrt{5}}{9+4\sqrt{5}}$ — ①
 $\frac{8-4\sqrt{5}}{4-5} - \frac{9+4\sqrt{5}}{81-80}$
 $-8 + 4\sqrt{5} - 9 - 4\sqrt{5} = -17$ Hence ratio

c) $\frac{x+1}{x+1} = x$ — ①
 $\frac{1}{x+1}$

d) In ΔPQR

i) $\cos \theta = \frac{15^2 + 7^2 - 13^2}{2 \times 15 \times 7}$ — ①

$= \frac{1}{2}$

$\therefore \cos \theta = \frac{1}{2}$

$\therefore \theta = 60^\circ$

$\angle SPQ = 147^\circ$

$\angle SPR = 87^\circ$ — ①

$A = \frac{1}{2} \times 9 \times 15 \times \sin 87^\circ$ — ①

$= 67.4074 \dots$ or 67.4

10

38 a) $x^2 + 12x + 36 = p + 36$ — ①
 $(x+6)^2 = p + 36$

$x+6 = \pm \sqrt{p+36}$

$x = -6 \pm \sqrt{p+36}$ — ①

For $p=25$, $x = -6 \pm \sqrt{25+36}$

$= -6 \pm 17$

$x = 11, -23$ — ①

b) $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$
 $1^3 = (x^3 + y^3) + 3xy(x+y)$

$1 = 19 + 3xy$

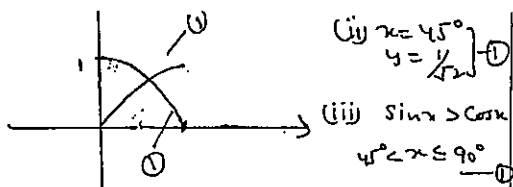
$3xy = -18$ — ①

$xy = -6$

$(x+y)^2 = x^2 + y^2 + 2xy$

$1 = x^2 + y^2 + 2x - 6$ — ①

$x^2 + y^2 = 13$



(ii) $\sin x > \cos x$
 $45^\circ < x < 90^\circ$

a) $\frac{5^{n+3} - 5^n}{5^n} = \frac{5^n \times 5^3 - 5^n}{5^n}$ — (1)

$\frac{5^n(5^3 - 1)}{5^n} = 124$ — (1)

e) $\frac{6(x^2 + 2)}{6(4 - x^2)}$ — (1)

$\frac{(x+2)(x^2 + 4 - 2x)}{(2-x)(2+x)}$ — (1)
 $= \frac{x^2 + 4 - 2x}{2-x}$

13

99 a) (i) $f(-2) = 1 - (-2)^2 = 1 - 2 = -1$ — (1)

(ii) $1 - x^2 = 2x + 1$
 $2x + x^2 = 0$
 $x(x+2) = 0$
 $x = 0, x = -2$ — (1)

(iii) $f(2x+1) = 1 - (2x+1)^2$ — (1)
 $= 1 - (4x^2 + 4x + 1)$
 $= 1 - 4x^2 - 4x - 1$
 $= -4(x^2 + x)$

b) In Δ $\cos \theta = \frac{CD}{BC}$
 $\therefore \cos \theta = \frac{CD}{BC}$
 In ΔABC , $\sec \theta = \frac{AC}{BC}$ — (1)
 $\sec \theta = \frac{AC}{BC}$
 $AC = CD + DA$
 But $DA = CD + 1$

$\therefore AC = 2CD + 1$ — (1)

$\sec \theta = 2 \cos \theta + 1$

(i) $\frac{1}{\cos \theta} = 2 \cos \theta + 1$ — (1)

$1 = 2 \cos^2 \theta + \cos \theta$ — (1)
 $2 \cos^2 \theta + \cos \theta - 1$

(ii) $2 \cos^2 \theta + \cos \theta - 1 = 0$
 $(2 \cos \theta - 1)(\cos \theta + 1) = 0$
 $2 \cos \theta - 1 = 0, \cos \theta + 1 = 0$
 $\cos \theta = \frac{1}{2}, \cos \theta = -1$
 but θ is an acute $\therefore \theta = 60$ — (1)

c) $y^2 + 16y + 64 + 94 - 6x$
 $(y+8)^2 + 30 - 6x$ — (1)
 $(y+8)^2 - 6(x-5)$
 $\therefore C = 8, B = -6, A = -5$ — (1)

d) $\frac{2}{3x-2c} + \frac{3}{2x-3c} = \frac{7}{2c}$

$\frac{4x - 6c + 9x - 6c}{6x^2 - 9xc - 4xc + 6c^2} = \frac{7}{2c}$

$\frac{13x - 12c}{6x^2 - 13xc + 6c^2} = \frac{7}{2c}$

$26xc - 24c^2 = 42x^2 - 91xc + 42c^2$
 $42x^2 - 117xc + 66c^2 = 0$ — (1)

$x = \frac{117c \pm \sqrt{(117c)^2 - 4 \times 42 \times 66c^2}}{84}$

$x = \frac{117c \pm \sqrt{2601c^2}}{84}$

$x = 2c, \frac{11}{4}c$ — (1)

13